

OpenManip: A Comprehensive Benchmark for Rigid and Articulated 6-DoF Manipulation

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Abstract—Imitation learning has emerged as a core paradigm in robotic manipulation, enabling agents to acquire complex behaviors from demonstrations. Yet, evaluating and comparing these methods—especially across tasks involving diverse object types and action spaces—remains challenging due to the lack of standardized benchmarks. In this paper, we introduce OpenManip, a benchmark and baseline suite for table-top open-instruction manipulation of both rigid and articulated objects. The dataset includes over 200 rigid and 150 articulated objects, with more than 2,000 tasks spanning three tracks—rigid-body, articulated-object, and combination—designed to test an agent’s ability to predict object positions and rotations in both simulation and real-world settings.

We benchmark several representative models and find a consistent pattern: large vision-language models often underperform in simulation benchmarks, particularly on tasks requiring precise physical reasoning, yet exhibit stronger results in real-world environments. These findings highlight key limitations in current simulation fidelity and the importance of grounding evaluation in real-world performance.

I. INTRODUCTION

Imitation learning has emerged as a fundamental paradigm in robotic manipulation, enabling agents to learn complex behaviors from expert demonstrations without requiring explicit reward engineering. By observing and imitating human or scripted actions, robots can acquire versatile skills in environments with rich sensory input and high-dimensional action spaces. This approach has shown particular promise in tabletop manipulation tasks, where precise control and contextual understanding are crucial. Despite its success, the field still faces challenges such as inconsistent benchmarks, limited task diversity, and heterogeneous evaluation protocols—making it difficult to fairly compare methods and measure generalization across scenarios.

While certain models can handle articulated object operations—such as opening doors or drawers—there remains a gap in evaluating manipulation tasks that combine both pick-and-place and articulated object interaction. Existing benchmarks often isolate one category or focus on limited action spaces, making it difficult to assess a robot’s ability to perform complex, sequential, and compositional tasks.

To address these limitations, we introduce OpenManip, a comprehensive benchmark for evaluating open-instruction tabletop manipulation across both rigid and articulated objects. OpenManip features over 200 unique rigid and 150 articulated objects, with more than 2,000 tasks covering diverse scenarios such as placing an item inside a drawer and closing it afterward. The benchmark is divided into three sub-benchmarks—Position, Rotation, and 6-DoF—each designed to test specific dimensions of manipulation, including

translational, rotational, and combined control capabilities. All tasks are annotated with natural language instructions and are designed to evaluate both the spatial understanding and planning capabilities of an agent.

Unlike many prior works that treat rotation as secondary or implicit, OpenManip isolates and emphasizes it as a core challenge in manipulation. This focus is particularly critical in 6-DoF tasks, where agents must precisely control both position and orientation to complete real-world objectives. By structuring tasks to explicitly test rotational understanding—alongside translation and articulation—we aim to expose the limitations of current policy architectures in capturing fine-grained spatial dynamics.

To this end, we benchmark a suite of representative imitation learning approaches—Diffusion Policy, ACT, OpenVLA, Octo, and RDT—on OpenManip. Our experimental results reveal several important trends. First, while large models such as ACT and Diffusion Policy exhibit strong performance overall, particularly in rigid-body tasks, their success is far less consistent in complex articulated or sequential tasks. Second, models with weaker architectural biases for temporal or spatial abstraction—such as OpenVLA, RDT, and Octo—perform substantially worse, especially on tasks requiring articulation or multi-step reasoning. Notably, even the top-performing models demonstrate a clear drop in performance when transitioning from simple pick-and-place tasks to more realistic manipulation involving drawers, doors, and combined actions.

These findings highlight a persistent gap between simulated performance and real-world robustness. Despite the progress of large-scale models in vision-language understanding and end-to-end control [1], [5], our results indicate that open-instruction manipulation in 6-DoF settings remains an unsolved challenge, particularly in environments with articulation and sequential dependency. OpenManip thus serves as both a diagnostic tool and a stepping stone—encouraging the development of more general, grounded, and reliable robotic policies for the next generation of embodied agents.

II. RELATED WORK

A. Robotic Manipulation Approaches

Robotic Manipulation is a complex and enduring challenge in the field of embodied AI, where an agent must understand and follow instructions to move objects to specific locations. Initial efforts in this area, as seen in earlier studies [7], [8], [18], employed task-and-motion-planning (TAMP) techniques that depend on predefined action sequences and nearly perfect knowledge of the environment to plan trajectories. However,

TAMP methods are often computationally inefficient and lack scalability in more complex scenarios. To address these limitations, recent research has shifted towards learning-based approaches [6], [9], [15], [20], [25], which aim to enhance both generalizability and efficiency. These methods are typically trained within simulators like iGibson [18], Habitat [22], and ManipulaTHOR [6] and focus on either high-level planning [6], [20] or low-level action prediction [9], [25]. While these approaches perform well in simulated environments, they often struggle with the sim-to-real gap [15] when applied in the real world. Recently, advancements in large models with open-world understanding [3], [13], [17], such as large language models (LLMs) [3] and visual language models (VLMs) [13], [17], have been leveraged for real-world Robotic Manipulation tasks. Some of these methods [16] use carefully designed prompts to describe environments and query LLMs or VLMs for guidance before execution. Others [23] train LLMs on self-collected data to directly output low-level actions. Unlike previous approaches that primarily focus on object location, our method emphasizes both location and orientation, leading to a 6-DoF (Degrees of Freedom) robotics manipulation approach.

B. Robotics Manipulation Benchmarks

Evaluating robotic manipulation is particularly challenging due to the need for detailed annotations of ground truth placements. Current benchmarks [6], [22], [24], such as those using simulators like ReplicatorCAD [19] or hand-crafted scenes, rely on high-quality object reconstructions [2] and provide annotations of both initial and target locations, which are essential for assessing placement accuracy. While real-world benchmarks [14] exist and can evaluate baselines using physical robots, they often fail to replicate testing environments consistently for all baselines and are limited by time constraints, reducing the scope of their evaluations. In contrast to these existing benchmarks, our approach introduces a 6-DoF robotic manipulation benchmark, which not only evaluates the object’s location but also its orientation.

C. Imitation Learning and Vision-Language-Action Model

Large models trained on vast amounts of internet data have shown significant promise in high-level planning tasks. The recent development of Vision-Language Models (VLMs) further narrows the gap between visual perception and textual interpretation, enabling embodied agents to better understand the semantics of scenes and instructions. However, VLMs still struggle with comprehending low-level manipulations, limiting their ability to directly generate executable actions for robots. One potential solution is to use VLMs as evaluators, generating goal states in the form of images, which VLMs then assess [11]. However, large-scale evaluations face challenges related to excessive time consumption and inconsistent scoring metrics. Our method mitigates these issues by decomposing the 6-DoF problem into two separate 3-dimensional tasks. Another challenge arises from the loss of information when converting 3D states into 2D images. To address this, our approach incorporates simulation-assisted

modules that preprocess raw scene data, translating it into a format that VLMs can interpret more effectively for decision-making.

III. OPENMANIP BENCHMARK

A. OpenManip Benchmark

OpenManip contains 2287 tasks, built around a curated set of over 200 unique rigid objects and 150 articulated objects. Each task is framed as an open-instruction manipulation problem where the agent receives a natural language command (e.g., "Place the blue cup inside the second drawer and close it"). The tasks are organized into three main benchmark tracks, designed to test different manipulation capacities:

Rigid-Body Track: Tasks involve only rigid objects (e.g., cups, blocks, bottles) and require accurate 3D translation and orientation. These tasks test the agent’s spatial reasoning and low-level motor control in isolation from object articulation.

Articulated-Object Track: Tasks focus on the manipulation of articulated objects, such as drawers, doors, and boxes with lids. These tasks evaluate the agent’s ability to model and interact with movable parts and handle dynamic state changes in the environment.

Combination Track: Tasks require coordination between rigid object placement and articulated object control (e.g., "Open the cabinet, put the green block on the middle shelf, and close the cabinet"). These tasks represent real-world composite skills that necessitate sequencing, conditional reasoning, and temporal planning.

Each object in OpenManip is represented as a 3D asset with fine-grained annotations, including its category, affordances (e.g., graspable, openable), and articulation parameters where applicable (e.g., hinge axis, open limits). All objects are placed in a variety of spatial configurations and contexts to increase scene diversity and reduce overfitting. The articulated objects include single- and multi-joint mechanisms, such as drawers nested in cabinets and double-door setups.

Task instructions are written in natural language and vary in phrasing to test the model’s robustness to linguistic ambiguity and compositionality. We employ template-based instruction generation augmented by large language models to create diverse and realistic command formulations. For example, instructions may vary from "Put the red block into the box" to "Place the crimson cube inside the container and close the lid."

To increase task challenge and ecological validity, object colors, sizes, and initial poses are randomized within defined constraints. Additionally, distractor objects are added in many scenes to test grounding and selective attention. All tasks are implemented in a physics-based simulation environment using PyBullet, enabling reproducible experimentation and rich sensory feedback (RGB-D, 3D poses, joint states, etc.).

B. OpenManip Benchmark Protocol

We rigorously design a training and evaluation protocol to ensure a structured and reliable assessment of the policy’s performance. Given the training data, the policy learns to imitate the demonstrated behavior. For evaluation, we provide

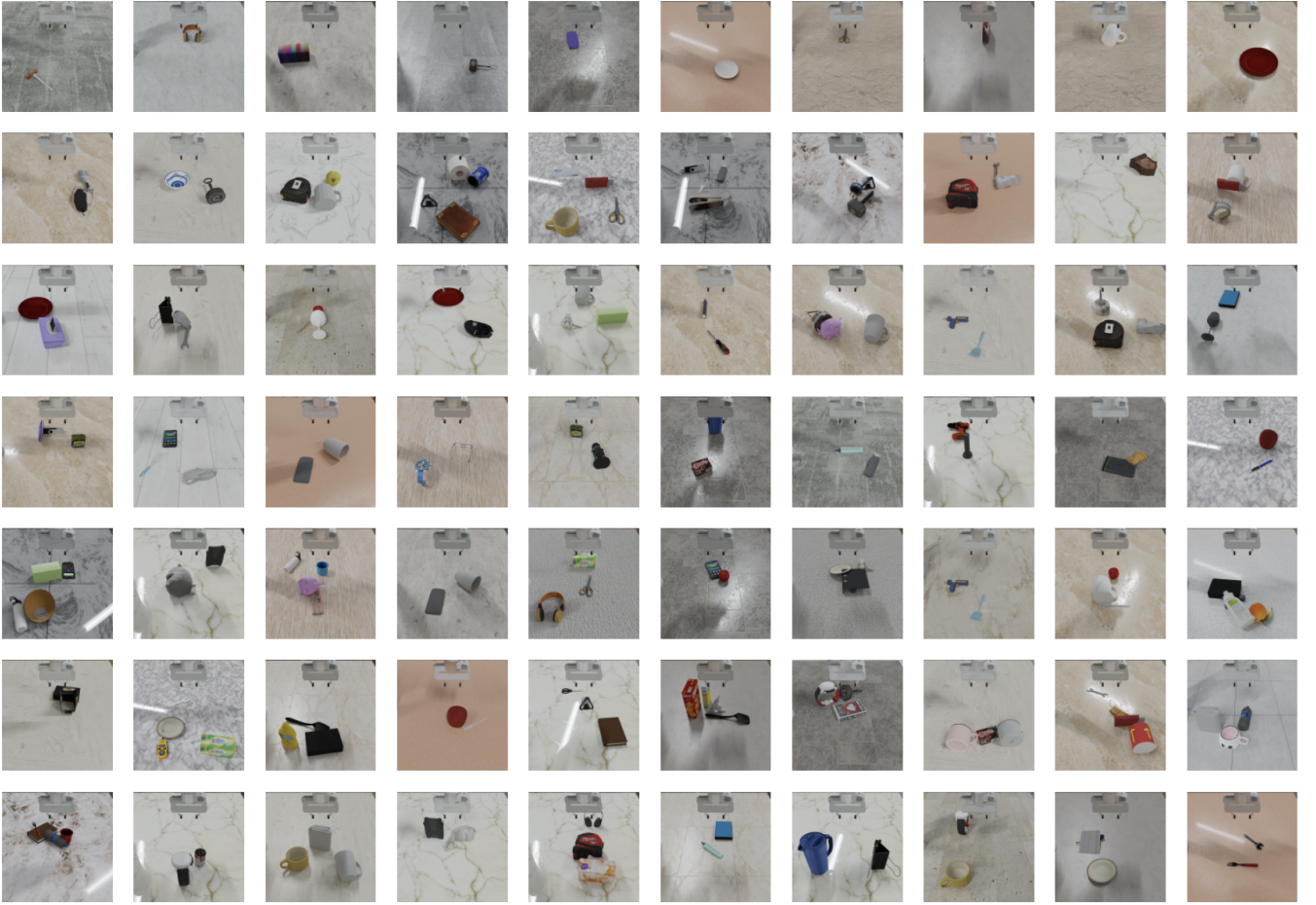


Fig. 1: **Simulation Benchmark and Dataset.** We set up a benchmark and dataset with more than 200 rigid and 150 articulated objects.

a standardized API that enables systematic assessment. As mentioned earlier, the training and evaluation follow a 9:1 ratio, ensuring that the policy is tested on novel scenarios not encountered during training.

To ensure consistency across methods, each task follows a standardized observation and action interface, with access to RGB-D inputs and 6-DoF control commands. The benchmark also defines a common success criterion based on object pose accuracy and final environment state (e.g., drawer closed, object placed correctly). Success is computed using thresholds on translation (5cm) and rotation (10 degrees), along with articulation completion where relevant.

For tasks requiring multiple steps, intermediate success is logged to analyze failure modes and evaluate partial completion. All results are reported as success rates, averaged across multiple seeds and randomized instances.

This benchmark aims not only to evaluate performance but also to expose generalization challenges across articulation, perception, and instruction understanding in a controlled and extensible platform.

IV. BENCHMARKING RESULTS AND ANALYSIS

V. RESULTS AND ANALYSIS

We evaluate five representative imitation learning methods on the OpenManip benchmark: **Diffusion Policy** [4], **ACT** [26], **OpenVLA** [12], **RDT** [12], and **Octo** [12].

These methods are tested across eight tasks spanning three tracks—*RigidBody*, *ArticulatedObject*, and *Combined*—to assess their ability to perform open-instruction 6-DoF manipulation.

A. Training Details

For specialist models, we train from scratch with action in 9-dim robot joint state space. Diffusion Policy [4] is implemented based on its original framework. We search several key hyperparameters, including observation and prediction length, to optimize performance for our tasks. ACT [26] is implemented with the original architecture and hyperparameters, except that the batch size has been increased to 512, with learning rate correspondingly enlarged to $1e-4$ to accelerate convergence. We train ACT on one A100 GPU for 2000 epochs and evaluate with the best checkpoints on the validation set.

For generalist models, the action is pre-processed into delta end-effector position space from absolute end-effector position space, and the gripper action is binarized to $\{0, +1\}$. Owing to the lack of time and resources, we are only able to fine-tune the generalist models in the single-task setting. For each task, OpenVLA kim24openvla is LoRA [10] fine-tuned (rank= 32) with 8 A100 GPU under official settings to convergence and reaches over 95% action token accuracy as proposed by kim24openvla during the training stage. During

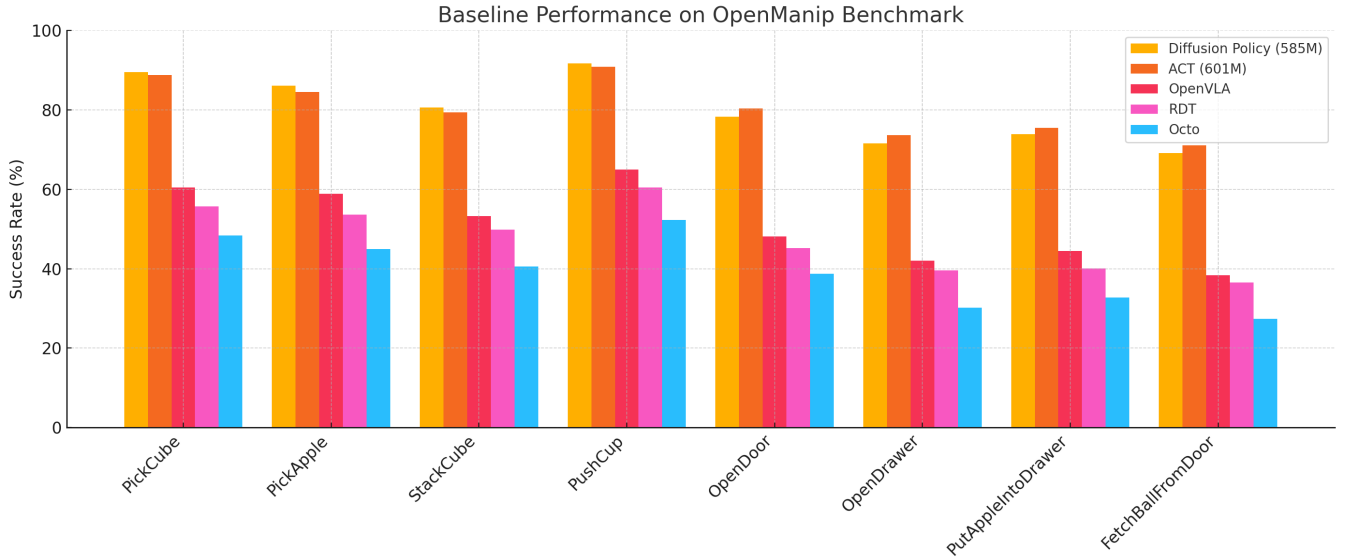


Fig. 2: Success rates (%) of baseline models on the OpenManip benchmark across different task categories. ACT and Diffusion Policy consistently outperform other methods, particularly on complex articulated and combined tasks.

evaluations, we employ Curobo [21] as the inverse-kinematics solver to transform the action to robot joint state space.

B. Benchmarking Results

Overall, results reveal a significant performance gap between top-performing models and others. Diffusion Policy and ACT consistently achieve high success rates across all tracks, particularly when scaled to larger model sizes. These two methods show robust generalization in both simple and complex manipulation settings, benefiting from their strong low-level control fidelity and temporal abstraction capabilities, respectively.

OpenVLA, RDT, and Octo demonstrate relatively weaker performance, particularly on articulated and multi-step combined tasks. While these models are able to handle basic pick-and-place scenarios, they struggle with sequential reasoning, articulation constraints, and long-horizon dependencies. Among them, Octo performs the worst overall, highlighting limitations in generalization and execution precision despite training on large-scale datasets.

A notable trend is the performance drop from rigid-body to articulated and combined tasks across all models. This underscores the increased complexity in tasks requiring object articulation and multi-phase planning. Even for high-performing models, tasks like opening a door, placing an object inside, and closing it present meaningful challenges, suggesting that current methods still fall short of true generalizable manipulation.

Interestingly, while large-scale models often show promise in real-world scenarios [1], the results on OpenManip suggest that simulation benchmarks remain a bottleneck. Some high-capacity models struggle in sim environments, pointing to a persistent sim-to-real gap and the importance of benchmark diversity and realism.

C. Benchmarking Conclusion

In summary, our benchmark analysis highlights three key insights: (1) strong imitation learning models can perform

well in open-instruction 6-DoF tasks when provided with the right architectural inductive biases; (2) articulation and sequential reasoning remain major challenges; and (3) model scaling improves performance but is not sufficient without corresponding improvements in structure and grounding. These findings reaffirm the need for standardized, challenging benchmarks like OpenManip to drive progress in open-world robotic manipulation.

VI. LIMITATIONS

While OpenManip provides a comprehensive benchmark for open-instruction manipulation, several limitations remain. First, although our task set spans both rigid and articulated objects with a diverse set of instructions, it still focuses primarily on single-tabletop settings with relatively constrained environments. This may limit generalization to more complex, multi-room or dynamic human-in-the-loop contexts.

Second, although our simulation setup strives for realism, the sim-to-real gap persists. Physical inaccuracies, visual domain gaps, and simplified contact dynamics may affect how well policies trained or evaluated in simulation transfer to real-world deployment. Although we include real-world testing for a subset of models, fully closing this gap remains an open challenge.

Third, our current evaluation emphasizes task completion and pose accuracy, but does not yet capture aspects such as manipulation smoothness, efficiency, or robustness under uncertainty. Future benchmarks could incorporate more nuanced metrics, including energy use, recovery from perturbation, or interpretability of failure modes.

Lastly, while we benchmark a range of popular imitation learning methods, the field is rapidly evolving. Incorporating more recent models, particularly those combining LLM-based planning with low-level control (e.g., multi-agent coordination or vision-action chaining), could offer deeper insights into generalization and reasoning capabilities.

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